

Non-Traditional Data Sources

The Hidden Treasure — taxonomy, quality, access and ethics for official statistics

Data Innovation For Africa (DI4A) · AfDB



Six sections, one thread



01

Why it matters

Pressures on NSOs and the mandate to innovate

8 min



02

Taxonomy

Six families of sources, what each can measure

15 min



03

Quality & bias

Adapted quality frameworks, the big-data paradox

12 min



04

Access & partnerships

Five access models and data-sharing agreements

8 min



05

Ethics & confidentiality

Law, re-identification risk, the Five Safes

7 min



06

Laboratory & way forward

Source Assessment Canvas, then report back

40 min

01

Why it matters

Why should an African statistical office care about data it never collected?



Demand is outpacing traditional collection

Four pressures converge on national statistical offices at the same time.

THE DEMAND SIDE

231

unique indicators in the global SDG framework — most to be disaggregated by sex, age, location and income

"Leave no one behind" means seeing small areas and small groups that national samples cannot reach.



Cost

Fieldwork, interviewers and follow-up absorb budgets that are flat or shrinking.



Timeliness

Censuses every ~10 years, surveys every 3–5: the decision is often taken before the figure exists.



Granularity

Samples designed for national totals cannot speak reliably about a district, a week or a niche group.



Fragility

Crises — epidemics, floods, conflict — interrupt fieldwork precisely when information is most needed.

Sources: UN Statistics Division, Global SDG Indicator Framework; typical census/survey cycles are indicative and vary by country.

The mandate to innovate is already written down

Principles, continental strategy and infrastructure all point the same way — with conditions.

2014

UN Fundamental Principles

Principle 5: data may come from all types of sources; choose on quality, timeliness, cost and respondent burden.

2018

SHaSA II adopted

AU Executive Council adopts the continental strategy, run through 18 Specialized Technical Groups.

2020

UN Big Data Hub, Kigali

Rwanda signs an MoU with the UN to host the Regional Hub for Big Data in Africa at NISR.

2025

STG17 Action Plan

First Annual Meeting (Kigali): Bureau, ToR and Action Plan 2025–2030 on big data, open data and AI.



The common condition: innovation is welcome only if quality, confidentiality and professional independence are preserved.

Sources: UNGA A/RES/68/261 (2014); STG17 workshop concept note (2026); UN-CEBD, Regional Hub for Big Data in Africa (unstats.un.org/bigdata).

Designed data versus found data

Neither is superior. They fail in opposite directions — which is exactly why they combine well.



Traditional (designed)



Non-traditional (found)

Origin	Designed for measurement	By-product of another activity
Concepts	Defined by the statistician	Defined by the data producer
Population	Known sampling frame	Unknown, self-selected users
Timeliness	Months to years	Hours to days
Granularity	Limited by sample size	Very fine — tile, hour, product
Marginal cost	High per observation	Low once access is secured
Continuity	Guaranteed by mandate	Depends on a third party
Main weakness	Cost, lag, non-response	Coverage bias, instability

02

Taxonomy

How do we name, sort and compare sources that were never meant to be statistics?



What makes a source "non-traditional"?



Working definition. Data generated as a by-product of an activity carried out for a purpose other than statistics — a transaction, a service, a sensor, a post — and repurposed to produce statistical evidence.

UNECE classification by generating process



Human-sourced

People record their experiences, often digitally.

Social media · search queries · reviews · crowdsourced maps · speed tests



Process-mediated

Records left by business or administrative processes.

Scanner & card data · mobile-money logs · call detail records · e-invoices



Machine-generated

Sensors and devices observing the physical world.

Satellite imagery · smart meters · GPS & AIS tracking · air-quality sensors



Quick test: if the record would still exist even if no statistician ever looked at it, the source is non-traditional.

Source: UNECE (2013), *Classification of Types of Big Data*, HLG-MOS Big Data Task Team.

The treasure map: six families of sources

Grouped by what an NSO actually works with. We open each chest on the next slides.



Lab: Day 4 (NTL)

1

Satellite imagery

Land, light, water, buildings



Lab: Day 3 (Ookla)

2

Crowdsourced measurement

Connectivity, maps, prices



3

Scanner, transaction & telecom

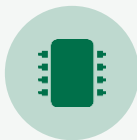
Prices, spending, mobility



4

Web scraping

Online prices, jobs, firms



5

Sensors & IoT

Energy, traffic, environment



6

Citizen-generated data

Local realities, voice, services

Satellite imagery and Earth observation



WHAT IT IS

Optical, radar and night-time light imagery from public missions (Sentinel, Landsat, VIIRS) and commercial providers — global, repeated, and largely free at medium resolution.

WHERE TO FIND IT

Copernicus Sentinel-2

Landsat 8/9

NASA Black Marble

Google Earth Engine

WHAT NSOs CAN MEASURE

- ✓ Land cover, crop area and crop condition
- ✓ Built-up area and settlement growth
- ✓ Night-time radiance: electrification, activity
- ✓ Floods, drought and water bodies
- ✓ Poverty mapping, combined with surveys

DOCUMENTED EXAMPLE

Jean et al. (Science, 2016) trained a model on daytime imagery and night lights to predict household consumption and assets in five African countries, including Rwanda.

⚠ WATCH OUT

The sensor sees a proxy, never the concept: light is not GDP, green is not yield. Calibrate against ground truth and revalidate over time.

10 m

Sentinel-2 resolution

~500 m

Black Marble grid (15")

1992–2013

DMSP-OLS archive

Sources: ESA Copernicus; NASA Black Marble (Román et al., 2018); Jean et al. (2016), Science 353(6301).

Crowdsourced measurement



WHAT IT IS

Observations contributed voluntarily by many users through an app or platform — network speed tests, collaborative maps, crowd-reported prices — then aggregated by the platform.

WHERE TO FIND IT

Ookla Speedtest open data

OpenStreetMap

Crowd price apps

WorldPop (auxiliary)

WHAT NSOs CAN MEASURE

- ✓ Download / upload speed and latency by tile
- ✓ Digital divide between regions and quarters
- ✓ Roads, buildings and facilities
- ✓ Market prices reported by volunteers
- ✓ Service availability at local level

DOCUMENTED EXAMPLE

Ookla publishes quarterly tiles of ~611 m (zoom 16) with avg_d_kbps, avg_lat_ms, tests and devices. Weighted by WorldPop population, they become a regional connectivity indicator — today's lab.

⚠ WATCH OUT

Tests come from people who choose to test, usually on smartphones and in towns. A tile with no test is not a tile with no signal. Check national coverage first.

~611 m

tile side at the equator

Quarterly

since 2019

CC BY-NC-SA

v4.0 licence: non-commercial, share-alike

Sources: Ookla, Global Fixed and Mobile Network Performance Maps (github.com/teamookla/ookla-open-data); WorldPop, Univ. of Southampton.

Scanner, transaction and telecom data



WHAT IT IS

Records produced by commercial processes: point-of-sale scanner files, card and mobile-money transactions, e-invoices, and mobile operators' call detail records (CDR).

WHERE TO FIND IT

Retailer scanner files

Mobile-money operators

E-invoicing (tax authority)

Operator CDR / signalling

WHAT NSOs CAN MEASURE

- ✓ Prices and quantities at product level (CPI)
- ✓ Retail turnover and consumption nowcasts
- ✓ Financial inclusion patterns
- ✓ Population presence and daily mobility
- ✓ Displacement after shocks

DOCUMENTED EXAMPLE

Ghana Statistical Service, Vodafone Ghana and Flowminder used aggregated CDR to publish mobility reports from April 2020, tracking adherence to COVID-19 restrictions by district.

⚠ WATCH OUT

Only formal, recorded activity is visible. Cash, informal trade and non-subscribers disappear; one operator's customers are not the population.

Early 2000s

first scanner-based CPIs (Europe)

2018

GSS–Vodafone–Flowminder agreement

Aggregated

only outputs leave the operator

Sources: UN Statistics Wiki, Mobile phone data guide, Appendix 1.5 Ghana; Flowminder (2020–2023); Blumenstock et al. (2015), Science 350(6264).

Web scraping and online platforms



WHAT IT IS

Automated collection of information displayed publicly online — product prices, job and property listings, company websites — plus platform and search-behaviour data.

WHERE TO FIND IT

E-commerce sites

Job portals

Property listings

Business directories

WHAT NSOs CAN MEASURE

- ✓ Online prices, updated daily
- ✓ Labour demand by skill and occupation
- ✓ Rents and housing supply
- ✓ Business registers and firm activity
- ✓ Early signals of price shocks

DOCUMENTED EXAMPLE

MIT's Billion Prices Project scraped daily online prices from retailers in dozens of countries and produced inflation series that tracked — and sometimes anticipated — official CPIs.

⚠ WATCH OUT

Websites change layout overnight; online offers are not transactions; urban, formal sellers dominate. "Publicly visible" is not "lawfully reusable" — check terms and law.

Daily

possible frequency

Robots.txt

and terms of use apply

Offer ≠ sale

the core proxy gap

Sources: Cavallo & Rigobon (2016), "The Billion Prices Project", *Journal of Economic Perspectives* 30(2); UNECE/ESS web intelligence work.

Sensors, IoT and telematics



WHAT IT IS

Fixed and mobile instruments that measure continuously: smart electricity and water meters, traffic counters, air-quality stations, GPS trackers, ship (AIS) and flight transponders.

WHERE TO FIND IT

Utility smart meters

AIS vessel tracking

Air-quality networks

Fleet GPS / telematics

WHAT NSOs CAN MEASURE

- ✓ Electricity and water consumption
- ✓ Port calls and maritime trade flows
- ✓ Traffic volumes and congestion
- ✓ Air and water quality exposure
- ✓ High-frequency activity indicators

DOCUMENTED EXAMPLE

AIS signals from ships are used internationally to nowcast port activity and trade volumes weeks before customs statistics are compiled — relevant for coastal economies.

⚠ WATCH OUT

Sensors drift, fail and move. A break in the series is often a hardware event, not an economic one: device metadata is part of the statistics.

Minutes

typical reporting interval

Device logs

needed to explain breaks

Utilities

natural first partners

Sources: UN Global Platform / UN-CEBD work on AIS data; UNECE (2014) quality framework, time-related and validity dimensions.

Citizen-generated data



WHAT IT IS

Data that people or their organisations deliberately produce to monitor issues that affect them — community surveys, participatory mapping, citizen science, service scorecards.

WHERE TO FIND IT

Civil-society monitoring

Citizen science projects

Participatory mapping

Community scorecards

WHAT NSOs CAN MEASURE

- ✓ Service delivery as experienced by users
- ✓ Local environmental conditions
- ✓ Groups absent from registers
- ✓ Gender-based and disability-related gaps
- ✓ SDG indicators with no other source

DOCUMENTED EXAMPLE

In Ghana, citizen-science observations on marine litter have been used to inform environmental policy and SDG reporting — an example cited around the Copenhagen Framework.

⚠ WATCH OUT

Powerful for voice and local detail, but rarely representative. Protocols, training and metadata determine whether it can enter official statistics.

2024

Copenhagen Framework welcomed (UNSC 55)

2025

v1.0 endorsed (UNSC 56)

Voice

as a data quality asset

Sources: Collaborative on Citizen Data, Copenhagen Framework on Citizen Data v1.0 (UNSC 56th session, 2025); IISD SDG Knowledge Hub (2024).

Comparing the six families at a glance

Indicative assessment for a typical African NSO — to be challenged with your own country experience.

	Timeliness	Granularity	Coverage in Africa	Ease of access	Privacy risk
 Satellite imagery	● ● ●	● ● ●	● ● ●	● ● ●	● ● ●
 Crowdsourced measurement	● ● ●	● ● ●	● ● ●	● ● ●	● ● ●
 Scanner, transaction & telecom	● ● ●	● ● ●	● ● ●	● ● ●	● ● ●
 Web scraping	● ● ●	● ● ●	● ● ●	● ● ●	● ● ●
 Sensors & IoT	● ● ●	● ● ●	● ● ●	● ● ●	● ● ●
 Citizen-generated data	● ● ●	● ● ●	● ● ●	● ● ●	● ● ●

Indicative, author's assessment for teaching purposes: ●●● = high, ●○○ = low. Coverage refers to the share of the population the source typically reflects.

03

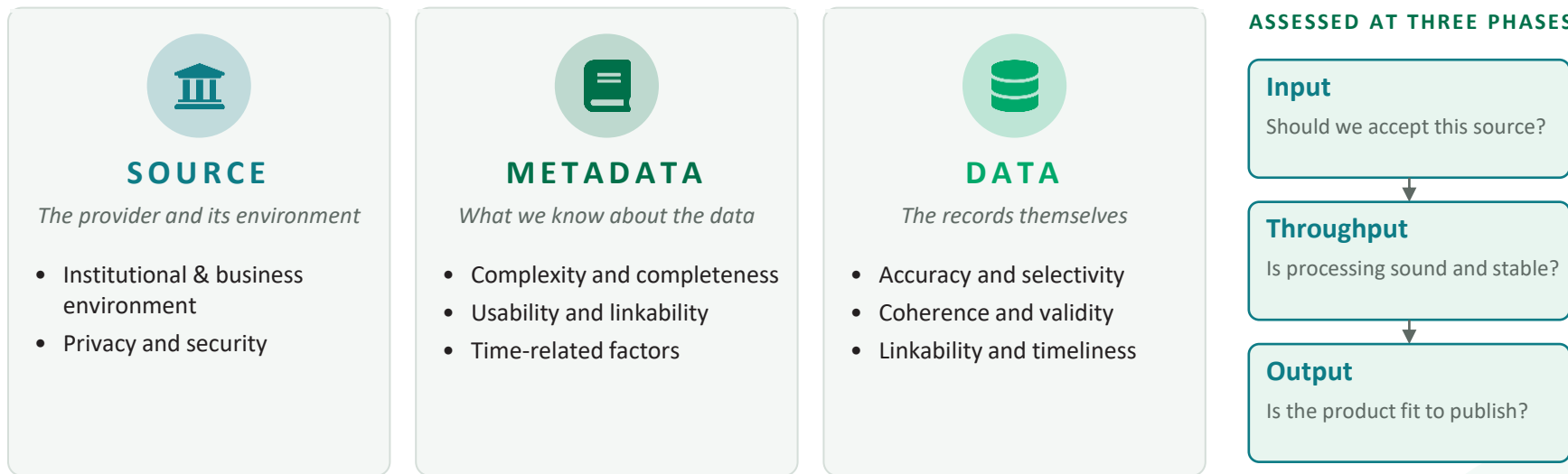
Quality & bias

If a source has ten million records, why can it still be wrong about everyone?



Extending quality frameworks to found data

UNECE proposes three "hyperdimensions", assessed at the input, throughput and output phases.

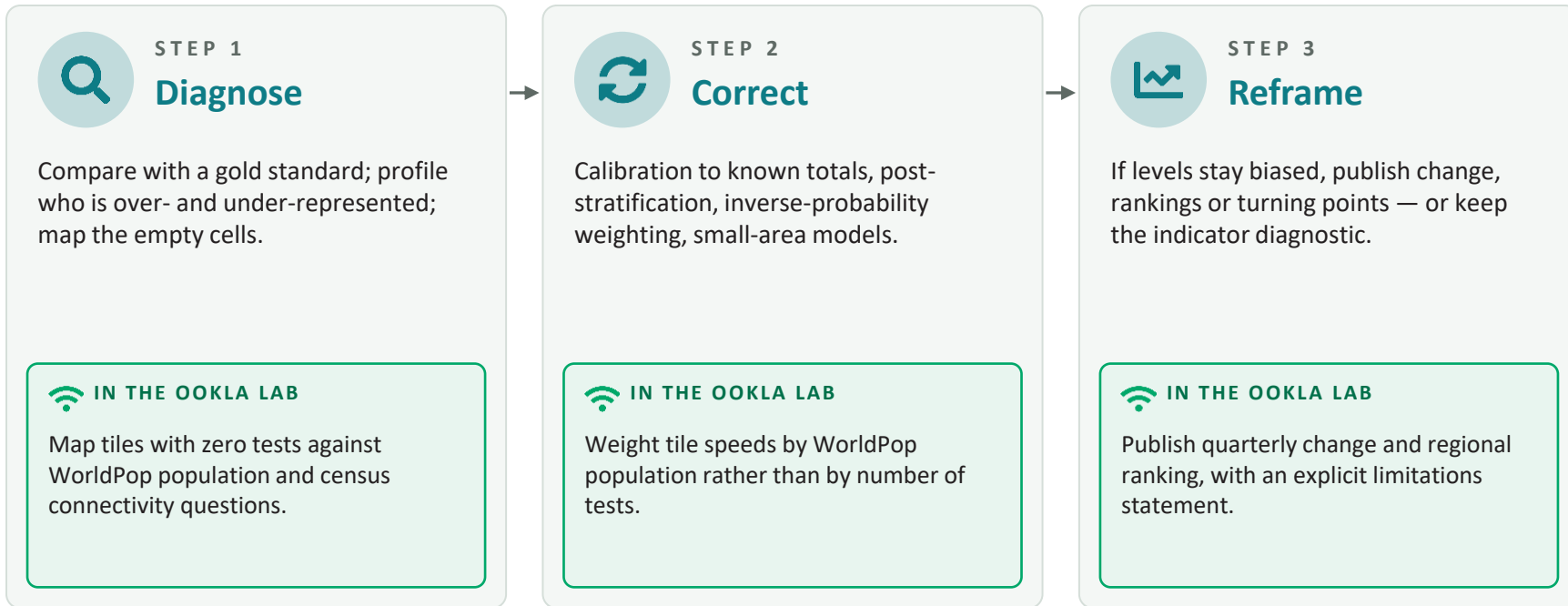


New questions classic frameworks never asked: Will this source still exist in three years? What is the fallback if the provider withdraws?

Source: UNECE (2014), *A Suggested Framework for the Quality of Big Data*, Big Data Quality Task Team (HLG-MOS).

Handling coverage bias: diagnose, correct, or reframe

Illustrated with the Ookla connectivity indicator you will build after the break.



Always name the concept, the proxy — and the gap

Validity problems are rarely technical. They are conceptual, and must be written down.

 TARGET CONCEPT	 PROXY OBSERVED	 THE GAP TO DOCUMENT
Local economic activity	→ Night-time light radiance	→ Saturation in cities, gas flares, unlit informal activity, agriculture
Internet access quality	→ Crowdsourced speed tests	→ A test is not a user; testers self-select; empty tiles ≠ no signal
Population present	→ Active SIM cards per antenna	→ Multi-SIM owners, shared phones, children and the unconnected
Consumer prices	→ Online advertised prices	→ An offer is not a transaction; informal markets are absent
Public opinion	→ Social-media sentiment	→ Opinion of active users, amplified by platform algorithms

04

Access & partnerships

The data exists. Why is getting it the hardest part of the project?



Five ways to obtain data held by others

From the fastest to start to the most durable. Most mature programmes combine several.

faster to start

more durable



Open data

Published under a licence: satellite archives, Ookla tiles, OpenStreetMap.

✓ Immediate

⚠ Licence limits



Purchase

Licensed from a vendor or aggregator for a defined use.

✓ Fast, clean

⚠ Cost, lock-in



Bilateral agreement

MoU with one firm: scope, outputs, safeguards, mutual benefit.

✓ Tailored

⚠ Slow to negotiate



Data collaborative

Several holders pool data for a public purpose via a trusted third party.

✓ Scale, neutrality

⚠ Governance load



Legal mandate

The statistics act obliges holders to supply data for official statistics.

✓ Durable

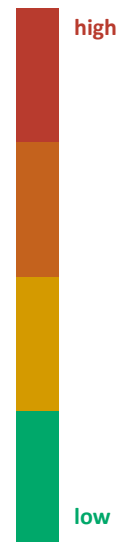
⚠ Slow, political

How the data moves matters as much as who owns it

Four technical access modes, from most to least exposure of individual records.

	Raw microdata transfer <i>Records copied to the NSO</i>	Maximum flexibility; highest disclosure risk; rarely accepted by providers.
	Pre-aggregated indicators <i>Provider computes agreed tables</i>	Simple and safe; method becomes a black box and hard to change.
	Code-to-data <i>NSO code runs behind the provider's firewall</i>	Only vetted aggregates leave. Reference model for telecom data.
	Privacy-enhancing tech <i>Enclaves, differential privacy, federated analysis</i>	Strong protection; needs advanced skills and infrastructure.

EXPOSURE OF
INDIVIDUAL RECORDS



RULE OF THUMB

Move the code to the data, not the data to the code.

Aggregate early, irreversibly, and only what the question needs.

Ten clauses every data-sharing agreement needs

A checklist to bring to the negotiating table — and to the STG17 data-governance reference guide.

1

Purpose

Statistical use only, stated precisely; no function creep.

2

Data specification

Fields, granularity, frequency, history and format.

3

Legal basis

Statistics act, data-protection law, consent where relevant.

4

Aggregation & thresholds

Minimum cell sizes; what may leave which environment.

5

Security

Storage, access control, audit trail, breach notification.

6

Roles & costs

Who does what; who pays; in-kind contributions.

7

Publication rights

Right to publish results and methods; provider review time.

8

Intellectual property

Ownership of raw data, code and derived indicators.

9

Duration & exit

Term, renewal, notice period, data deletion, fallback.

10

Governance

Steering committee, dispute resolution, periodic review.

05

Ethics & confidentiality

Technically possible, legally allowed, and publicly acceptable — are all three true?



Four layers of rules apply at the same time

An NSO must check each layer before any record moves.



GLOBAL

UN Fundamental Principles of Official Statistics

Principle 5: any source, chosen on quality, timeliness, cost and burden.
Principle 6: individual data strictly confidential.



CONTINENTAL

African Charter on Statistics · Malabo Convention

Professional independence and confidentiality; AU data-protection treaty in force since 8 June 2023.



NATIONAL

Statistics act · Data-protection law

Access powers for statistical purposes; lawful basis; supervisory authority; sector rules (telecom, banking).



CONTRACTUAL

Agreements · Terms of service · Licences

e.g. Ookla open data under CC BY-NC-SA 4.0: attribution, non-commercial, share-alike.

Sources: UNGA A/RES/68/261 (2014); African Charter on Statistics (2009); AU Malabo Convention (2014, in force 2023); Ookla licence.

Anonymisation is more fragile than it looks

UNIQUE IN THE CROWD

4

location–time points

were enough to single out

95%

of 1.5 million people in a mobile-phone mobility dataset — with no names in it.



Individual re-identification

Rare combinations of place and time betray a person, even without names or numbers.



Group harm

Aggregates can still expose a community — displaced people, minorities, informal settlements.



Function creep

Data shared for statistics must never feed surveillance, enforcement or commercial targeting.



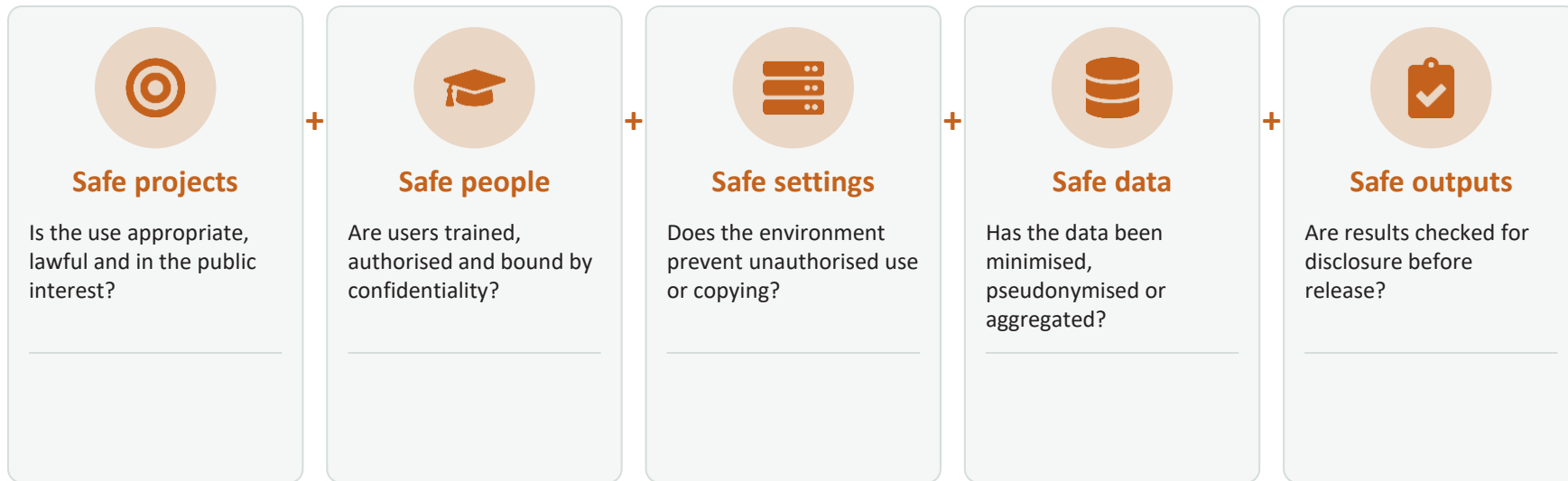
Mosaic effect

Harmless tables, once combined with other public data, can reveal what each one hid.

Source: de Montjoye, Y.-A., Hidalgo, C., Verleysen, M. & Blondel, V. (2013), "Unique in the Crowd", *Scientific Reports* 3: 1376.

The Five Safes: a practical frame for safe access

Risk is managed by combining five controls — no single one is sufficient.



 Controls compensate each other: a richer dataset may be acceptable only in a safer setting, with trained people and output checking.

Source: Desai, T., Ritchie, F. & Welpton, R. (2016), "Five Safes: designing data access for research", UWE Economics Working Paper 1601.

Ethics questions to ask before, during and after

Short enough to use in every project review; strong enough to stop a bad one.

BEFORE

Design & access

- ? Is there a clear public-interest purpose?
- ? What is the lawful basis, and who approved it?
- ? Would data subjects find this use reasonable?
- ? Is a less intrusive source available?

DURING

Processing

- ? Is data minimised and aggregated as early as possible?
- ? Who can see what, where, and is it logged?
- ? Are models documented and reproducible?
- ? Are small cells suppressed?

AFTER

Publication & use

- ? Are limitations published as prominently as results?
- ? Could an output harm a group or community?
- ? Who becomes invisible in this data?
- ? When will raw inputs be deleted?

Glossary of essential terms

Non-traditional data

Data generated for a non-statistical purpose and repurposed as evidence.

Coverage bias

Error arising when the units in the data differ systematically from the target population.

Calibration

Adjusting weights so that estimates reproduce known totals from a trusted source.

Call detail record (CDR)

An operator log of a phone event: antenna, time, duration.

Code-to-data

Access mode where the analyst's code runs in the provider's environment.

Proxy

An observable variable standing in for a concept that cannot be measured directly.

Data defect correlation (ρ)

Correlation between being recorded and the value measured; zero under random sampling.

Small-area estimation

Model-based estimates for fine geography, combining survey and auxiliary data.

Quadkey

A string identifying a web-Mercator map tile; used as the Ookla tile ID.

Five Safes

Projects, people, settings, data and outputs — a frame for safe data access.

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DISCUSSION

Thank you

*The treasure was never the data. It is the decisions
the data makes possible — safely.*

